

Impact of the implementation of Madrid's low emission zone on NO₂ concentration using Sentinel-5P/TROPOMI data

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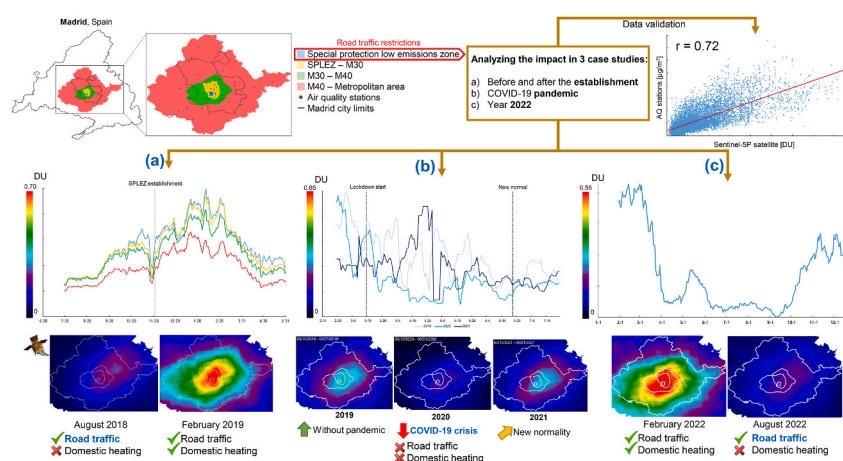
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HIGHLIGHTS

- Study based on NO₂ emissions in Madrid's low emissions zone (LEZ) using TROPOMI satellite data.
- Peak concentrations of traffic-related NO₂ in the city accumulate outside the SPLEZ.
- Heating seems to be a main contributor to increasing NO₂ emissions.
- NO₂ emissions decreased during the strict lockdown period of COVID-19 pandemic.
- The comparison of satellite data with surface-level measurements showed a high correlation.

GRAPHICAL ABSTRACT



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ABSTRACT

In recent years, several measures and policies have been introduced to safeguard healthy NO₂ levels in urban areas, including the creation of low emission zones (LEZs). In November 2018, a traffic-restricted area known as the Special Protection Low Emission Zone (SPLEZ) was established in the core of the city of Madrid, embracing most of its central district. This study aims to evaluate whether the implementation of the LEZ has led to improvements in air quality by analyzing NO₂ concentrations in various case scenarios. For this purpose, data from the Tropospheric Monitoring Instrument (TROPOMI) onboard the Sentinel-5 precursor (Sentinel-5P) were used. The pre- and post-LEZ implementation seasonal analysis (from July 2018 to May 2019) and the entirety of 2022 indicated that the epicenter of transport-related NO₂ concentrations did not lie within its boundaries. However, NO₂ levels within the SPLEZ peaked during winter, with heating and meteorological factors playing a significant role. Additionally, the findings from the case study of the COVID-19 pandemic demonstrated the influence of

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human activities on the NO₂ concentration in both the SPLEZ and the entire metropolitan area. NO₂ emissions decreased during the strict lockdown period and steadily increased as activities resumed. Satellite data was compared to surface-level measurements in each case study, resulting in a linear correlation coefficient of 0.72. TROPOMI data was found to closely align with surface-level observations as pollution levels increased, contrasting with better air quality scenarios. Despite certain limitations, it can serve as a valuable tool for providing complementary information to surface-level data, particularly in characterizing changes in pollution distribution patterns.

1. Introduction

Nitrogen monoxide (NO) and nitrogen dioxide (NO₂) are the most widely studied nitrogen oxides (NO_x) in the scientific literature due to their impacts on human health. NO_x are produced under high temperatures and solar radiation conditions. They can be generated naturally (i. e. lightning, volcanic eruptions) or in anthropogenic processes (i.e. combustion of fossil fuels, biomass burning). Among the anthropogenic sources, domestic heating (4.4%), power generation (18.4%), and road transportation (26.3%) are the most notable contributors globally (McDuffie et al., 2020). Most atmospheric NO_x emissions rapidly transform into NO₂, which can react with other substances to form secondary pollutants: particulate matter, nitric acid (HNO₃), peroxyacyl nitrates (PANs), and ozone (O₃).

Elevated levels of atmospheric NO₂ can have adverse effects on human health, particularly on the respiratory, immune, and circulatory systems (Buzikov et al., 2017; WHO, 2013; Shaw and Van Heyst, 2022). From an environmental perspective, it can cause acidification and eutrophication of ecosystems, as well as disruptions to vegetation and water quality (Emberson et al., 2001; Hakeem et al., 2017). Even in urban areas, NO₂ can damage buildings and historical heritage (Cicek et al., 2009).

For these reasons, recent years have witnessed efforts to develop air quality improvement plans in major European cities. One leading measure is the creation of Low Emission Zones (LEZs), where activities such as combustion-based transportation are restricted to maintain pollution levels under healthier margins (Ku et al., 2020). Additionally, transforming urban environments into wider and more open spaces favors walking and electric transportation, as well as micro-mobility and the creation of new green infrastructures.

Madrid is an area of particular interest in this regard. Previous studies have found that the metropolitan area of Madrid has the highest mortality in Europe due to NO₂ pollution, with an average annual level of 39.2 µg/m³ (Khomenko et al., 2021). The current limit set by the European Union is 40 µg/m³ (European Union, 2008), which will be halved with new regulations (European Union, 2022). Furthermore, the World Health Organization (WHO) recommends 10 µg/m³ in its latest air quality guidelines (WHO, 2021). To address this issue, the creation of the first low-emission zone in the city of Madrid was approved in 2018. It was located in the central district, where NO₂ limits were frequently exceeded. Earlier studies have reported that the establishment of LEZs can lead to improvements in air quality, as evidenced by ground-based measurements (Panteliadis et al., 2014; Qadir et al., 2013; Santos et al., 2019). While not all European cities have achieved equally effective results (Ku et al., 2020; Ma et al., 2021), traffic restrictions, particularly on diesel vehicles, generally result in a reduction of pollution and other potential benefits such as improved public health and road safety (Settey et al., 2019). Previous research has demonstrated the effectiveness of the SPLEZ through the exclusive use of data from Madrid's air quality stations network (Salas et al., 2021). To date, this represents the first endeavor to examine the changes brought about by the establishment of the SPLEZ in Madrid, employing additional new technologies such as remote sensing.

The first satellite data on atmospheric NO₂ were obtained in 1995 through the Global Ozone Monitoring Experiment (GOME, Burrows et al., 1999). Subsequent projects, such as the Scanning Imaging

Absorption Spectrometer for Atmospheric Cartography (SCIAMACHY, Bovensmann et al., 1999) and the Ozone Monitoring Instrument (OMI, Levelt et al., 2006), kept improving the quality of the acquired images. Nowadays, the Tropospheric Monitoring Instrument (TROPOMI) is the spectroscopic sensor that has provided the highest spatial and temporal resolution ever achieved at a global level. The device is onboard the Sentinel-5 Precursor satellite (Sentinel-5P), which is part of the European Union's Copernicus program for monitoring the Earth's surface, atmosphere, and oceans. While many environmental and meteorological factors influence the distribution of NO₂ in the atmosphere, it has been demonstrated that atmospheric column measurements obtained by TROPOMI exhibit a strong correlation with surface measurements, especially in urban environments (Cersosimo et al., 2020; Goldberg et al., 2021; Jeong and Hong, 2021). This has also been observed in the case of Madrid (Morillas et al., 2024).

Studying how remote sensing information can complement or even substitute data obtained through ground-based air quality networks is of great interest. This is particularly relevant because satellite images can provide a regional perspective on the distribution of pollution over time and are not limited to specific geographical points. Understanding the main patterns resulting from implementing a LEZ can be especially useful in areas with high and persistent pollution or with limited surface level air quality networks.

To study the impact under different scenarios, the implementation of the SPLEZ was evaluated in three parts. The first part involved comparing NO₂ levels before and after its establishment. The second part analyzed NO₂ concentration over the SPLEZ and its surroundings under an exceptional and critical event, using the COVID-19 pandemic as the most recent example. The third one focused on observing patterns throughout an entire calendar year. The findings were complemented by analyzing territorial variables (transportation, heating, population density) and meteorological factors (wind speed, wind direction) associated with pollution distribution in the city. The whole study was complemented and compared with data collected by air quality monitoring stations in Madrid.

2. Material and methods

2.1. Study area

The region of interest in this work is the SPLEZ in the city of Madrid. Madrid is located in the center of the Community of Madrid (see Fig. 1). The SPLEZ, originally named Madrid Central, was a 472-ha area within the perimeter shown in Fig. 1d. The establishment of Madrid Central aimed to promote pedestrianization, traffic reduction, and, consequently, lower pollution levels, particularly nitrogen oxides (NO_x). Since its initial implementation on November 30th 2018, it has undergone several modifications in terms of restrictions and extension, following changes in the central administration.

The main mobility restrictions targeted vehicles owned by non-residents in the area, based on their labeling (Ministerio de la Presidencia, Relaciones con las Cortes y Memoria Democrática MPRCMD, 2018). Electric vehicles (Zero label) and hybrids or those powered by natural gas (ECO label) were allowed to circulate and park freely, with ECO vehicles limited to a maximum parking duration of 2 h. Diesel and gasoline vehicles, depending on their year of registration (B and C

labels), could only enter for specific purposes with special access permission, provided they searched for a public or private parking space. Lastly, vehicles without a label were not allowed to enter (City of Madrid, 2022a).

At the end of May 2019, successive changes were introduced, leading to the current situation of 2023. The SPLEZ was renamed Distrito Centro as part of the Madrid 360 project (City of Madrid, 2022b). Greater freedom of movement was established, especially for C-labeled vehicles with high occupancy. Residents and businesses were equally placed regarding regulations, although progressive restrictions are expected in the coming years (City of Madrid, 2022a). The perimeter of the north-west area was reduced until Mártires de Alcalá Street, and unrestricted circulation was allowed through Bailén Street. Due to its high pollution levels, the Plaza Elíptica area was added (Fig. 1d), although it was not considered in this study.

In the first case analysis, three additional regions were considered to further evaluate pollution patterns, as shown in Fig. 1c. Given the close relationship between road transportation and NO₂ emissions in urban environments, the following areas were taken into account: the region from the SPLEZ to the M30 (the city's major ring road); the area from the M30 to the M40 (the subsequent ring road); and the zone stretching from the M40 to the rest of the metropolitan area, where the highest traffic influx between the city and outer municipalities occurs.

2.2. Study period

The first part of the study assesses the impact of the SPLEZ on NO₂ concentration before and after November 30th 2018. To maintain consistency regarding the aforementioned restrictions, and due to limitations in TROPOMI's data availability (see Section 2.3.), the study period was set between June 28th 2018, and May 31st 2019.

The second part of the study examines the effects of the COVID-19 crisis on the SPLEZ. On March 14th 2020, the Spanish government declared a state of alarm (MPRCMD, 2020a). The situation was consecutively extended until June 21st of the same year, with the adoption of a "new normality" (MPRCMD, 2020b). Starting from March 15th, a period of home confinement took place, leading to the suspension of all daily activities. On May 2nd, a phased de-escalation process began, and certain activities were gradually resumed. Subsequently, during the new normality, some containment measures were kept on or reintroduced, although not as strict as the initial lockdown. To provide a pre- and post-perspective, the study period for the second case spans from February 14th to July 22nd for the years 2019, 2020, and 2021. This allowed for a year-on-year comparison as well.

The third part analyzes the NO₂ concentration over the SPLEZ during a full calendar year. The most recent one, 2022, was selected, covering the period from January 1st to December 31st.

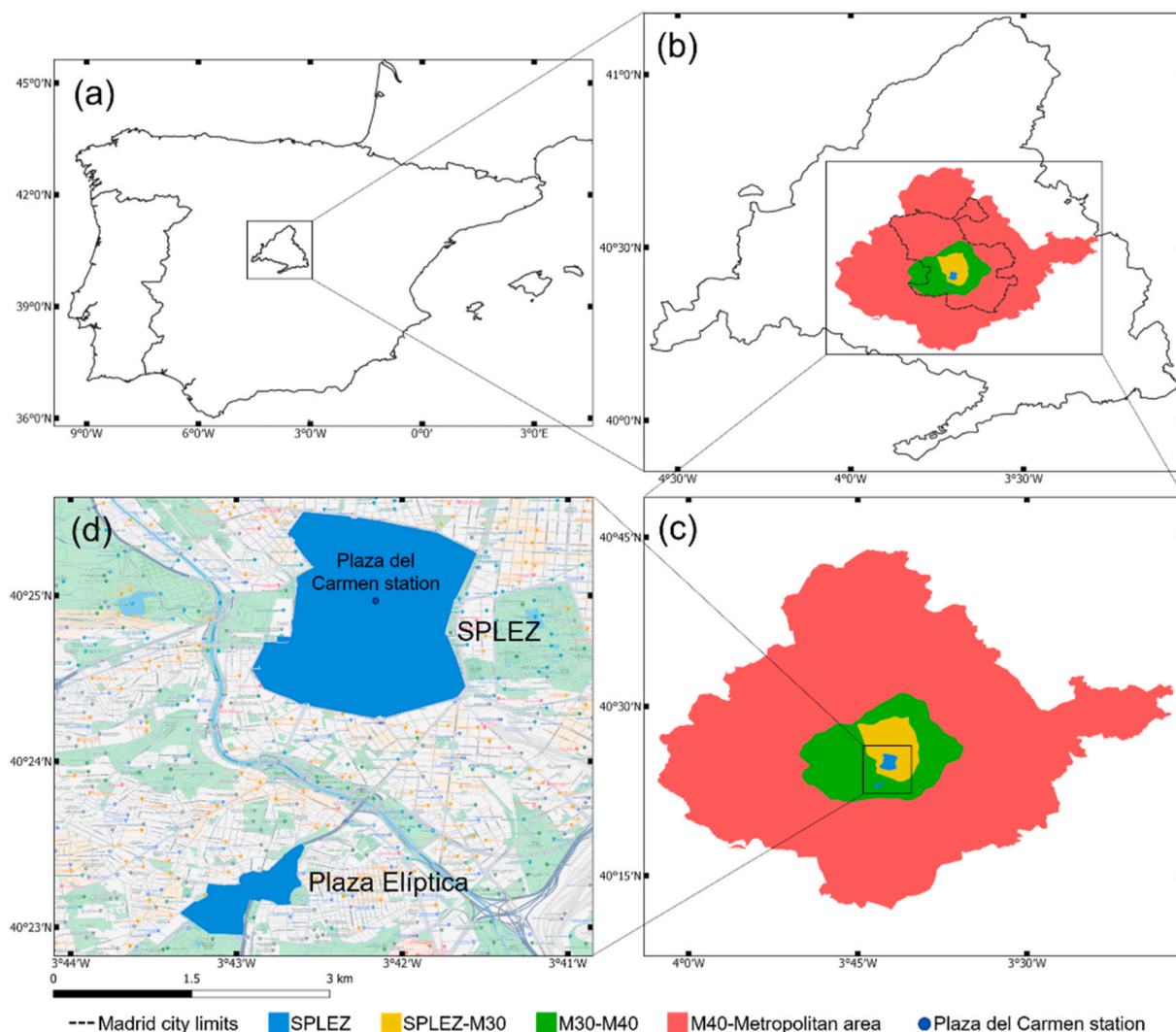


Fig. 1. Location of the Community (a) and the city (b) of Madrid; study areas (c) and the low emission zones (d).

2.3. Satellite NO₂ data

TROPOMI, the payload on the Copernicus Sentinel-5P satellite, was launched in October 2017 and began providing data on June 28th 2018. Its primary objective is to monitor the atmosphere for climate change and air quality purposes. It is expected to remain operational until the future launches of Sentinel-4 and Sentinel-5. Sentinel-5P describes a near-polar, sun-synchronous orbit with an altitude of 824 km. It crosses the equator at 13:30 Mean Local Solar Time for 14 orbits per day, providing a global daily view of the atmospheric composition. In the case of Madrid, overpass times always fall between 12:00 and 4:00 p.m. Local Time (UTC+1 in winter and UTC+2 in summer) and were estimated based on the start of granule time (Eskes et al., 2022). Occasionally, two observations of the same point within an approximate 100-min difference occur when two consecutive orbits overlap on certain days. In such cases, average daily values were calculated. The initial 3.5×7.5 km² spatial resolution was improved to 3.5×5.5 km² in August 2019.

TROPOMI is a passive optical sensor designed to measure the total column amounts of several trace gases across the UV and SWIR spectral regions (Van Geffen et al., 2019). Data products are classified into three processing levels. Level-0 corresponds to raw data; Level-1 products provide geo-located top of the atmosphere Earth radiances for each spectral band; and most Level-2 products are derived using a Differential Optical Absorption Spectroscopy (DOAS) algorithm. This process retrieves tropospheric and stratospheric vertical column densities (VCDs), similar to other satellite missions (Eskes and Eichmann, 2018; Williams et al., 2017). Level-2 products include information on aerosols and other substances, but this study focuses solely on NO₂. Products are available in two versions: near real-time (NRT) and offline (OFFL). NRT products are accessible within 3 h of sensing, while OFFL products used in this study offer higher data quality but require longer processing times (Eskes and Eichmann, 2018).

Level-3 images, referred to hereafter, result from an adapted reprocessing system in OMI (Ozone Monitoring Instrument) and are openly available. They are derived from Level-2 products by using the harp-converter drive tool with the bin-spatial operation. The data is merged into a $0.01 \times 0.01^\circ$ resolution mosaic, producing an orthorectified raster layer (Google Earth Engine, 2023). This collection of images has also undergone masking to exclude points with high cloud cover or low-quality pixels (tropospheric_NO2_column_number_density_validity < 50). VCDs lower than -0.001 mol/m² were filtered out following recommendations. Level-3 NO₂ data are provided in mol/m² and were converted to Dobson units (DU).

2.4. Software tools

All image retrieval and processing were conducted using Google Earth Engine (GEE), a cloud-based platform that allows access to and operations on extensive catalogs of imagery on a global scale (Google Earth Engine, 2023).

The maps were created using QGIS, a free and open-source Geographic Information System software that allows the import and manipulation of images obtained from GEE (QGIS, 2023).

2.5. In-situ data

In each case study, satellite data were consistently compared with in situ measurements taken by the air quality monitoring network of the Community of Madrid and the City of Madrid. The whole region is equipped with 48 stations, with 24 located within the city. These stations measure various substances, including SO₂, CO, PM_{2.5}, PM₁₀, O₃, NO, and NO₂, at the surface level (City of Madrid, 2023a; Community of Madrid, 2023). NO₂ levels are registered by chemiluminescence analyzers every hour, and datasets are provided in mass concentration (µg/m³). Data undergo a validation process that is published alongside

each corresponding value. Records from Plaza del Carmen station were particularly crucial in this study, as it is the only one located within the SPLEZ limits. Fig. 2 shows all stations in the Community of Madrid and their distribution according to the four regions of interest mentioned in Section 2.1 (see also Fig. S1). A linear correlation analysis was conducted by comparing average data between 12:00 and 4:00 p.m. from TROPOMI and ground stations, following the approach of previous studies (Morillas et al., 2024). In situ information was provided by the air quality monitoring network of both the city and the Community of Madrid for every case study considered. For Section 3.2, data between June 28th 2018, and May 31st 2019 from all stations falling within the metropolitan area were used. For Section 3.3, data from Plaza del Carmen station between February 14th and July 22nd for the years 2019, 2020, and 2021 were used. For Section 3.4, data from the Plaza del Carmen station for the whole year 2022 were employed.

2.6. Territorial and meteorological factors

To provide a more in-depth analysis of the spatial distribution of NO₂, its relationship with various variables—both territorial and meteorological—has been studied. This includes official traffic data for each of the study zones (City of Madrid, 2023b), Copernicus Atmospheric Monitoring Service (CAMS) data to assess the ratio between NO_x emissions from heating and road traffic within the SPLEZ (Granier et al., 2019), official population density data to allocate demand from domestic heating (MITERD, 2021), and ERA5-Land hourly wind speed and wind direction data to study atmospheric transport dynamics (Muñoz Sabater, 2019).

3. Results

3.1. Data comparison

To ensure correspondence between satellite and in-situ data, both sources were compared. The resulting linear correlation coefficient obtained was $r = 0.72$, as shown in Fig. 3. Regions with higher and more persistent pollution (typically urban areas) exhibited stronger values than those with cleaner air conditions (mostly rural sites located within the furthest region, Table S1).

3.2. Air quality in the SPLEZ before and after implementation

To perform the comparison before and after the establishment of the SPLEZ in Madrid, the daily average of Level-3 tropospheric TROPOMI NO₂ concentration in the whole area between 12:00 and 4:00 p.m. from June 28th 2018 to May 31st 2019 was obtained. To assess the potential influence and spillover effects on its surroundings, this analysis was extended to the three additional regions. The study period was divided into two parts, with the implementation date of the SPLEZ (November 30th 2018) serving as a benchmark.

To ensure a direct collocation between Sentinel-5P and in-situ measurements, satellite data were precisely obtained at the locations of the air quality stations. Due to the variable availability of high-quality TROPOMI images, comparisons were limited to days when both satellite and in-situ values were recorded simultaneously. This is complemented by mean values measured by TROPOMI for the entire areas before and after November 30th, 2018, as shown in Table 1. The daily mean NO₂ concentration observed by TROPOMI in the whole study period decreased with increasing distance from the SPLEZ. Fig. 4a displays the evolution of the 30-days moving average of NO₂ concentration as seen by TROPOMI, while Fig. 4b shows measurements from the air quality network. Before the implementation of the SPLEZ, a general upward trend in daily NO₂ concentration was detected, followed by a downward trend afterward, indicative of seasonal variations. Differences between the study areas tended to expand during periods of greater pollution but contracted in intervals of better air quality. The sensitivity to these

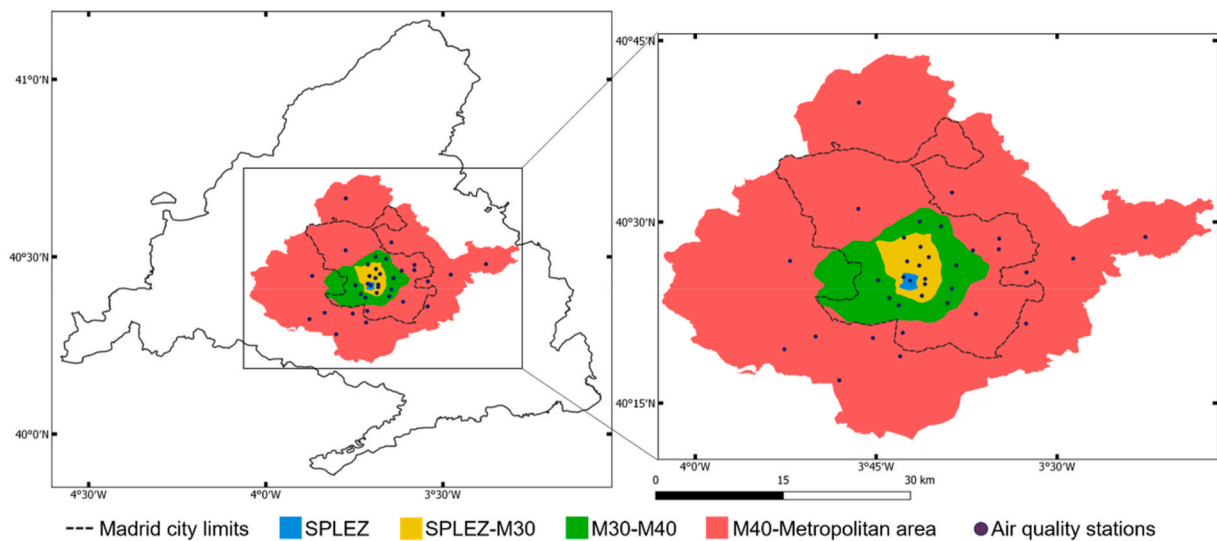


Fig. 2. Spatial distribution of the air quality stations network of the Community and the City of Madrid.

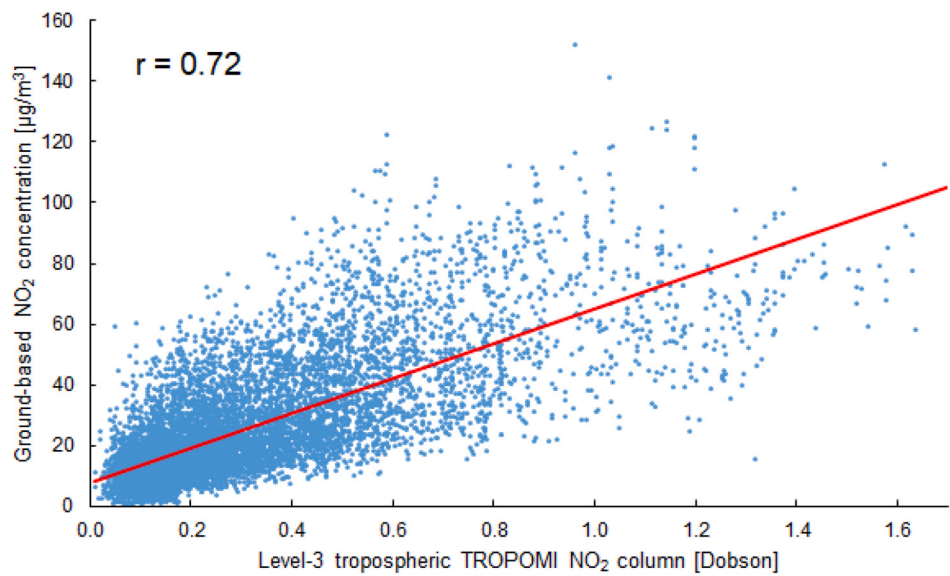


Fig. 3. Scatterplot of Level-3 tropospheric TROPOMI NO₂ vs. in situ observations averaged between 12:00 and 4:00 p.m. for every case study.

Table 1
Average Level-3 tropospheric TROPOMI NO₂ concentration (DU) over the four study areas before and after the SPLEZ establishment.

Period	SPLEZ	SPLEZ-M30	M30-M40	M40-Metropolitan area
06/28/2018 - 11/29/2018	0.301	0.300	0.271	0.191
11/30/2018 - 05/31/2019	0.431	0.418	0.375	0.246

changes increased as proximity to the SPLEZ became closer. Additionally, there was a sudden reduction in NO₂ concentration in the days before the SPLEZ implementation, followed by a common gradual enhancement until a decline at the end of February.

Fig. 5 displays changes in traffic intensity compared to the average levels of 2015. The 12-month moving average shows a noticeable decrease from early 2018, with a milder effect on the surrounding areas. This decline continued after the implementation of the SPLEZ, but rose again following its closure in June 2019.

NO₂ spatial patterns were also analyzed to explore significant behaviors in its distribution. As shown in Fig. 6, the average NO₂ concentration was low in July (Fig. 6a) and August (Fig. 6b). A gradual increase took place in September (Fig. 6c), except for November (Fig. 6e), and reached a maximum in February (Fig. 6h). Peak NO₂ concentrations were identified in the northeast area of the SPLEZ-M30 region, notably in November (Fig. 6e) and December (Fig. 6f). The highest NO₂ values were recorded in January (Fig. 6g) and February (Fig. 6h), with the SPLEZ and the M30 region exhibiting the most significant pollution levels. Subsequently, concentrations gradually declined until the end of May (Fig. 6k). It was also observed that, during warm months, peak concentrations generally shifted to other regions outside the center but gathered around the SPLEZ in colder periods.

This is consistent with CAMS data (see Fig. S2), which depict a gradual decrease in NO_x emissions from road traffic in the SPLEZ, a circumstance that can be extrapolated to the rest of the city. Residential emissions have remained constant for years. In winter, emissions from heating occasionally surpass those from traffic, while the greatest difference occurs during the summer. The most polluted areas in winter correspond to the most densely populated municipalities in the region

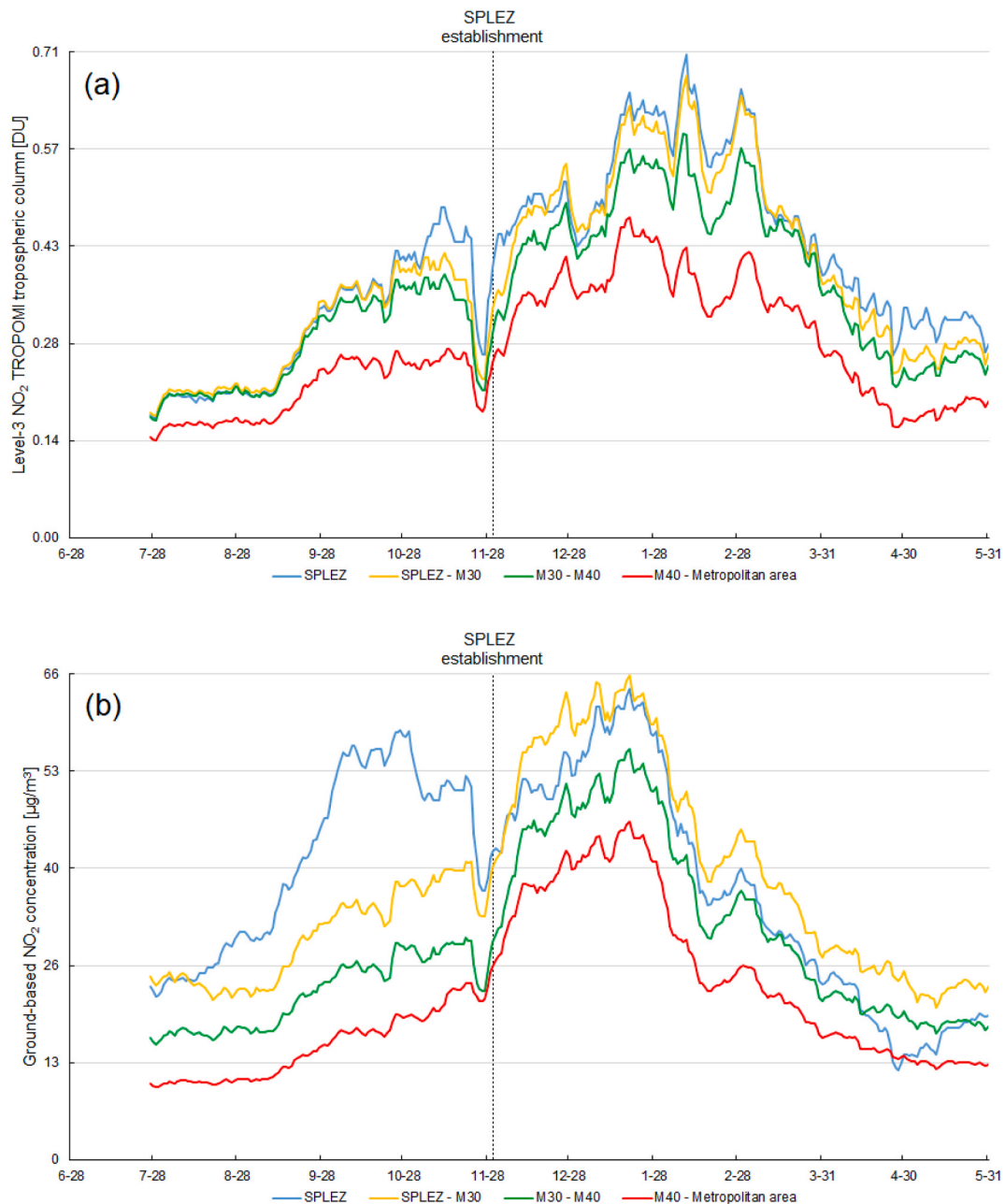


Fig. 4. 30-day moving average of (a) Level-3 tropospheric TROPOMI NO₂ concentration (DU) and (b) in situ NO₂ concentration (µg/m³) between 12:00 and 4:00 p.m. at each station's specific location within the four study areas.

(see Fig. S3).

The results on the average wind speed and direction are presented in Fig. 7. It was observed that higher wind speeds directly influenced the dispersion of NO₂ in the SPLEZ during the study period.

Additionally, a closer relationship between the wind and the distribution of NO₂ concentration is noticeable during periods with higher speeds and more persistent directions (see Fig. 8). This is particularly noticeable in July 2018 (Fig. 8a), March 2019 (Fig. 8i) or April 2019 (Fig. 8j), where both higher speeds (see also Fig. 7b) and consistently northeastern directions tend to follow the distribution pattern of NO₂ in relation to the city center.

3.3. SPLEZ before and after the COVID-19 pandemic crisis

To analyze the impact of the COVID-19 pandemic on the SPLEZ, an examination of the NO₂ concentration was conducted, considering three

consecutive time intervals from February 14th to July 22nd. These periods were delineated by the measures adopted by the national government during the first stages of the crisis: the start of the lockdown (March 15th) and the end of the first state of alarm (June 21st). The analysis was done for the years 2019 (pre-pandemic), 2020 (during the pandemic), and 2021 (post-pandemic) to facilitate a year-on-year comparison.

Table 2 presents the mean values of the NO₂ column measured by TROPOMI in the three periods considered. A general decreasing trend is observed for the years 2019 and 2021. In 2020, the average concentration decreased abruptly during the state of alarm (from 0.46 to 0.14 DU) and slightly increased again after adopting the new normality. In 2021, a lower value was observed during the first period compared to previous years.

The temporal evolution is shown in Fig. 9 where, given the duration of the study period, a 10-day moving average was considered to

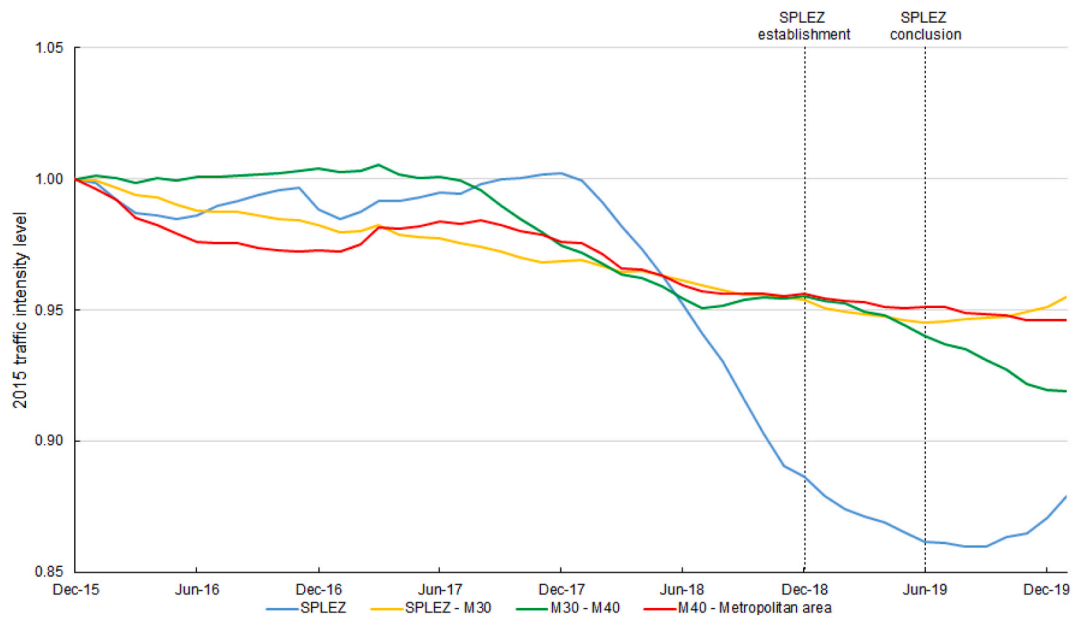


Fig. 5. 12-month moving average of weekday traffic intensity in each study area normalized to the average levels of 2015. Prepared by the authors based on the data provided by the City of Madrid.

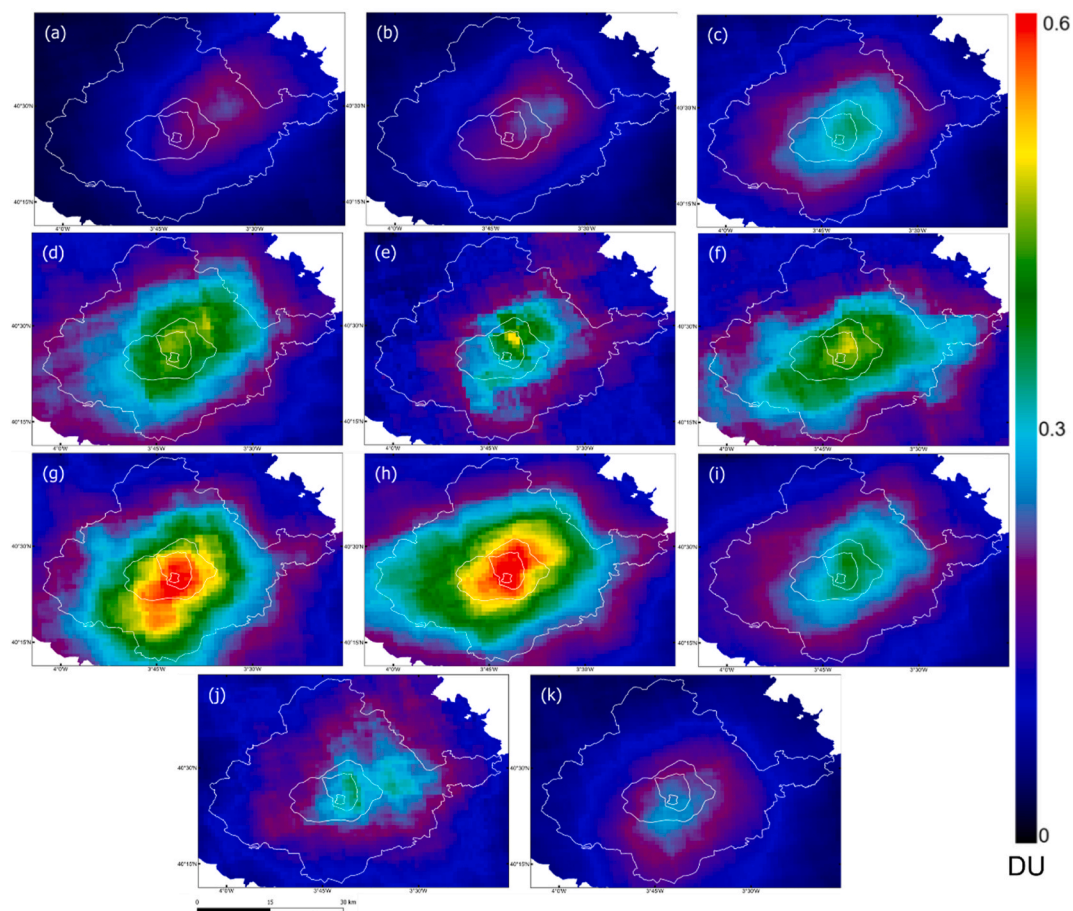


Fig. 6. Monthly maps of mean Level-3 tropospheric TROPOMI NO_2 column (DU) in (a) July 2018 (including the last 3 days of June), (b) August 2018, (c) September 2018, (d) October 2018, (e) November 2018, (f) December 2018, (g) January 2019, (h) February 2019, (i) March 2019, (j) April 2019 and (k) May 2019. The white lines from the inside to the outside represent the SPLEZ and the surrounding areas.

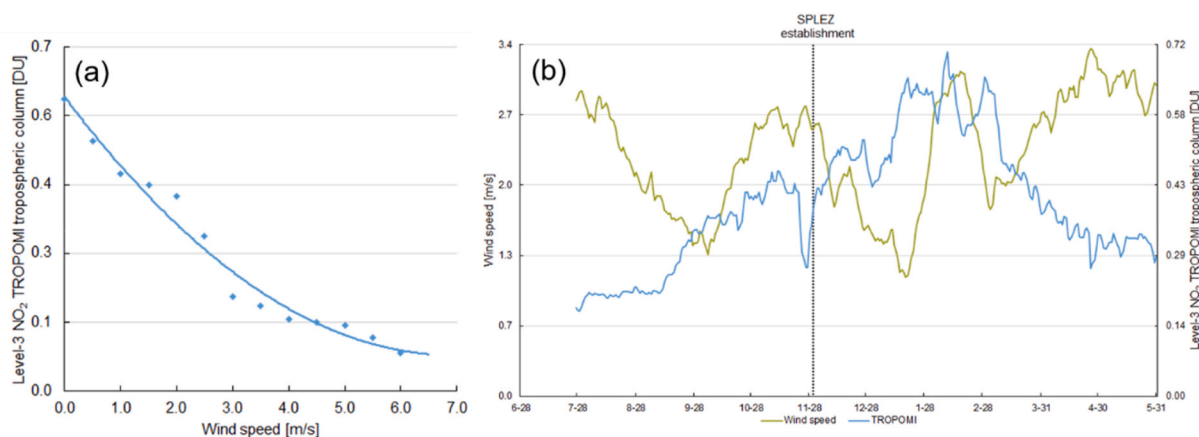


Fig. 7. Average speed of the 10 m wind (m/s) and Level-3 TROPOMI NO₂ tropospheric column (DU) between 12:00 and 4:00 p.m. in the SPLEZ during the study period. (a) Presents NO₂ concentration based on wind speed; (b) displays the 30-day moving average of wind speed and NO₂ concentration.

characterize the trends. Fig. 9a and b illustrate the contrast between satellite and in situ measurements, following the same approach as in the first case. Both datasets exhibit a comparable trend. In 2020, there was an immediate decrease in the NO₂ concentration in the SPLEZ with the start of the lockdown. In early May of that year, coinciding with the beginning of the lockdown de-escalation phases, there was an enhancement that aligned the trends again until the early stages of the new normality period.

Overall, the highest daily mean values were measured in the year 2019, and the lowest, in 2020. The year 2021 shows a similar evolution to 2019 but with generally lower concentrations, especially in the first period.

The spatial distribution of NO₂ was addressed by obtaining a mean map for 2019, 2020, and 2021 comprising each study period, as shown in Fig. 10. Peak values were generally centered towards northeastern regions outside the SPLEZ, especially in warmer periods. During the state of alarm of 2020 (Fig. 10e), anthropogenic emissions were barely distinguishable, although a slight rebound was noticeable after the establishment of the new normality (Fig. 10f). Except for colder periods (Fig. 10a and d), the SPLEZ did not particularly exhibit the highest NO₂ concentration compared to its surroundings.

3.4. Annual NO₂ pollution over the SPLEZ in 2022

Data for the whole year 2022 were gathered to evaluate the spatio-temporal evolution of NO₂ pollution in the SPLEZ as observed by TROPOMI throughout a natural year. Table 3 shows mean monthly values in the area. Higher concentrations were found in the colder months, especially in January with 0.50 DU and February with 0.55 DU, and lower values in warmer periods, particularly in summer months.

Fig. 11 displays the average daily values and the 30-day moving average in 2022 as seen by TROPOMI and Plaza del Carmen ground station. NO₂ pollution clearly decreased at the beginning of March and remained stable until October, with the exception of a temporary rise in May.

Monthly maps were also created to address spatial patterns, as shown in Fig. 12. TROPOMI NO₂ images revealed the SPLEZ as one of the NO₂ pollution hotspots during colder months (Fig. 12a and b). However, peak concentrations shifted to northeastern regions in months of lower pollution (i.e. spring, summer).

4. Discussion

The main objective of analyzing the impact of the SPLEZ before and after its establishment was to verify potential changes in NO₂ emission patterns detected by TROPOMI and characterize them. Additionally,

similar to previous research using in situ data (Salas et al., 2021; Santos et al., 2019), the effects of the SPLEZ in adjacent areas were analyzed. In such studies, the temporal progression of NO₂ concentration before November 30th 2018 in every region of interest followed a similar qualitative trend, showing higher accumulation in central areas compared to the outskirts, while maintaining consistent quantitative differences between the study zones. After the implementation date, air quality monitoring stations found an exclusive downward trend in the SPLEZ until the end of May 2019 and no remarkable negative spillover effects in terms of NO₂ pollution.

This study has confirmed that TROPOMI can successfully detect these behaviors, although with some exceptions. Firstly, the differences in NO₂ concentration are greater as the measured values increase and are lower in scenarios with better air quality. This can be observed in two consecutive timeframes in Fig. 4: from November 30th until March 1st, and from then until May 31st. As found in previous studies (Goldberg et al., 2021; Jeong and Hong, 2021), the sensor's resolution still limits the distinction of spatial concentration gradients at this scale. While these changes are clearly observed in the in situ data (refer to Fig. 4b), TROPOMI appears to be more responsive to concentration changes in scenarios with higher pollution (as evidenced in the period between December 2018 and January 2019, where TROPOMI NO₂ values for the SPLEZ are below those of its surroundings). This could be attributed to the fact that the spectral behavior of the gas is more easily detectable by the sensor at increased concentrations. Nevertheless, its spatiotemporal distribution follows a consistent pattern in all cases.

It is important to note that this study does not compare observations of the same nature (surface concentration vs tropospheric column). Although NO₂ is generated and accumulates mostly in urban environments, TROPOMI observes higher layers of the troposphere (Huang et al., 2023), potentially detecting emissions from sources located in different areas. Despite the strong satellite-in situ data correlation, remote sensing observations are subject to environmental factors that play a crucial role in NO₂ dynamics. Complex orographic and atmospheric conditions can give rise to a range of air quality scenarios. Although Madrid does not present many heterogeneous landforms, it is set on a mountain basin that can prevent movements between air layers in thermal inversion episodes (Su et al., 2008). Besides, NO₂ accumulates beneath the planetary boundary layer (PBL), which expands in the morning and comprises at night (Su et al., 2018), determining its vertical distribution. Temperature can impact all the above-mentioned aspects and additionally favor NO₂ conversion back to NO, generating ozone as a side pollutant (Gorai et al., 2015). Previous studies have indicated that the NO₂ lifetime in the atmosphere is longer in winter than in summer (Shah et al., 2020). However, this was not further addressed in this paper and should be the focus of future research.

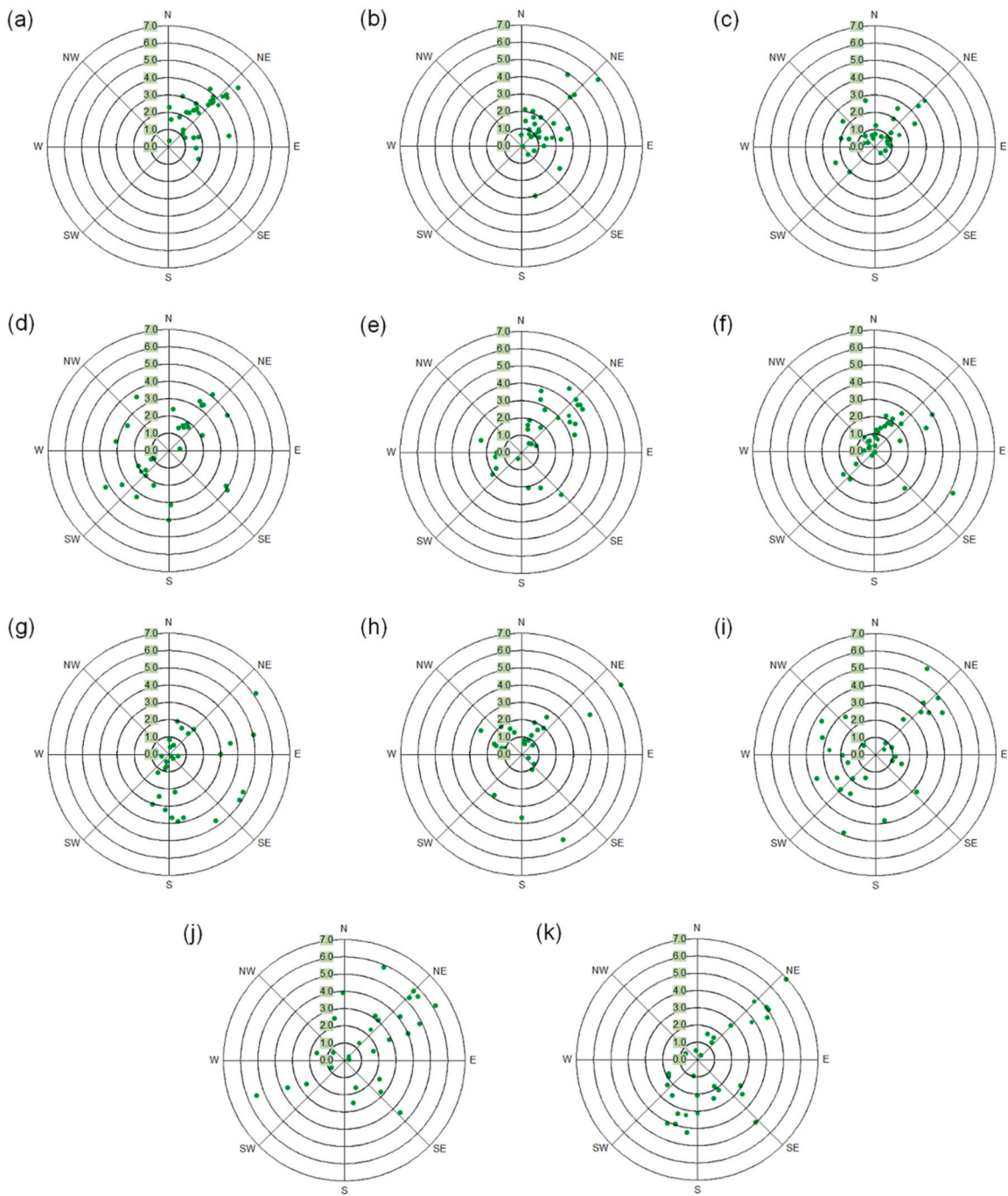


Fig. 8. Daily average speed of the 10 m wind (m/s) and wind direction in Madrid between 12:00 and 4:00 p.m. in (a) July 2018 (including the last 3 days of June), (b) August 2018, (c) September 2018, (d) October 2018, (e) November 2018, (f) December 2018, (g) January 2019, (h) February 2019, (i) March 2019, (j) April 2019 and (k) May 2019. Prepared by the authors based on ERA5-Land hourly data.

Table 2
Average Level-3 tropospheric TROPOMI NO₂ concentration (DU) in the SPLEZ during the three periods considered in 2019, 2020, and 2021.

Period	2019	2020	2021
02/14–03/14	0.45	0.46	0.29
03/15–06/21	0.29	0.14	0.24
06/22–07/22	0.22	0.17	0.18

Among these factors, wind speed is arguably the most influential, as it can rapidly disperse pollution away from its sources (Contreras and Ferri, 2016). A comparison between Figs. 6 and 8 unveils an intriguing relationship in months characterized by high average wind speeds and persistent wind directions, where corresponding spatial distributions are evident. Wind plays a pivotal role in determining the distribution of the NO₂ tropospheric column and hinder the perception of changes that are clearly observed at the surface level (see also Fig. 7b). The temporal evolution observed by TROPOMI (Fig. 4) shows the usual seasonal trend of NO₂ pollution, with an increase in colder months and a decrease during the middle part of the year (see also Fig. 11). Since

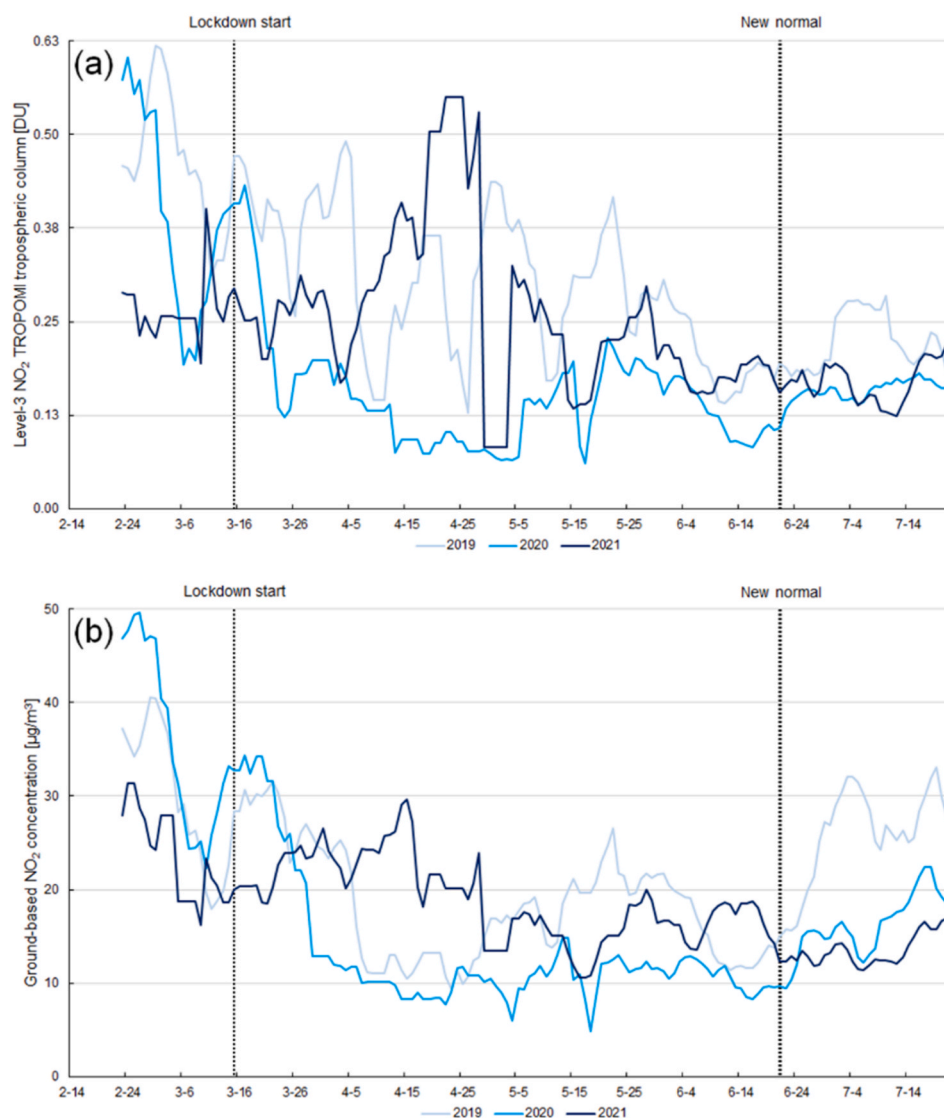


Fig. 9. 10-day moving average of (a) Level-3 tropospheric TROPOMI NO₂ concentration (DU) and (b) in situ NO₂ concentration (µg/m³) at Plaza del Carmen station location between 12:00 and 4:00 p.m. from February 14th to July 22nd in 2019, 2020 and 2021.

Sentinel-5P data was not available until June 28th 2018, a qualitative comparison of interannual pollution patterns to glean further insights into the impact of the SPLEZ was not possible. The sudden decrease in concentrations before November 30th 2018 was common to all areas considered, and it may have been due to beneficial climatic conditions (Salas et al., 2021). NO₂ has an atmospheric lifetime of hours, which is further shortened during periods of clarity and sunlight that promote photolysis processes and chemical conversion (Williams et al., 2017). The increase in NO₂ concentration observed since early 2018 could be attributed to the execution of construction works to remodel Gran Vía, the primary roadway crossing Madrid Central. This might have impacted traffic flow, leading to congestion and delays before the implementation of the SPLEZ.

Monthly maps in Fig. 6 reveal that the SPLEZ experiences elevated NO₂ concentrations and is situated closer to the epicenter of emissions during colder months, when the demand for heating fuel rises (see Fig. S3). In warmer periods, the distribution shifts, and higher values move towards peripheral areas, primarily to the northeast. These patterns suggest that, during winter, heating rather than transportation might be the predominant source of emissions within the SPLEZ, as illustrated in Fig. S2. In spring and summer, transportation becomes the primary source of NO₂, highlighting the SPLEZ-M30 region as the main

concentration hotspot, while the SPLEZ itself is not as prominent.

The study on the impact of the COVID-19 pandemic on the SPLEZ revealed NO₂ as an accurate tracker of human activities. Table 2 shows that the average NO₂ concentration significantly dropped in 2020 during the state of alarm (from 0.46 DU to 0.14 DU), representing only 48% of the same period of 2019 (0.29 DU). Fig. 9 illustrates the substantial impact of these restrictions on pollution over the SPLEZ, mostly related to mobility limitations. The data indicates a drastic decrease in NO₂ levels during the lockdown (Bauwens et al., 2020; Tobías et al., 2020; Virghileanu et al., 2020), followed by a subsequent rise with the resumption of certain activities during the de-escalation phase. Both the satellite and the air quality stations equally detected these changes, although TROPOMI data revealed differences between study zones more distinctly at higher pollution scenarios (see comparison in Fig. 9). Gradually, the trend in 2020 aligned with those of 2019 and 2021, especially after the start of the new normality. In 2021, it was noticed that the concentration of NO₂ did not follow the same trend during the pre-pandemic period compared to previous years, showing a 37% lower concentration. While various factors could contribute to this deviation, it is worth noting that some restrictions remained in place for several months. Many activities have not fully resumed to this day, and others, such as remote work, have permanently changed due to the impact of

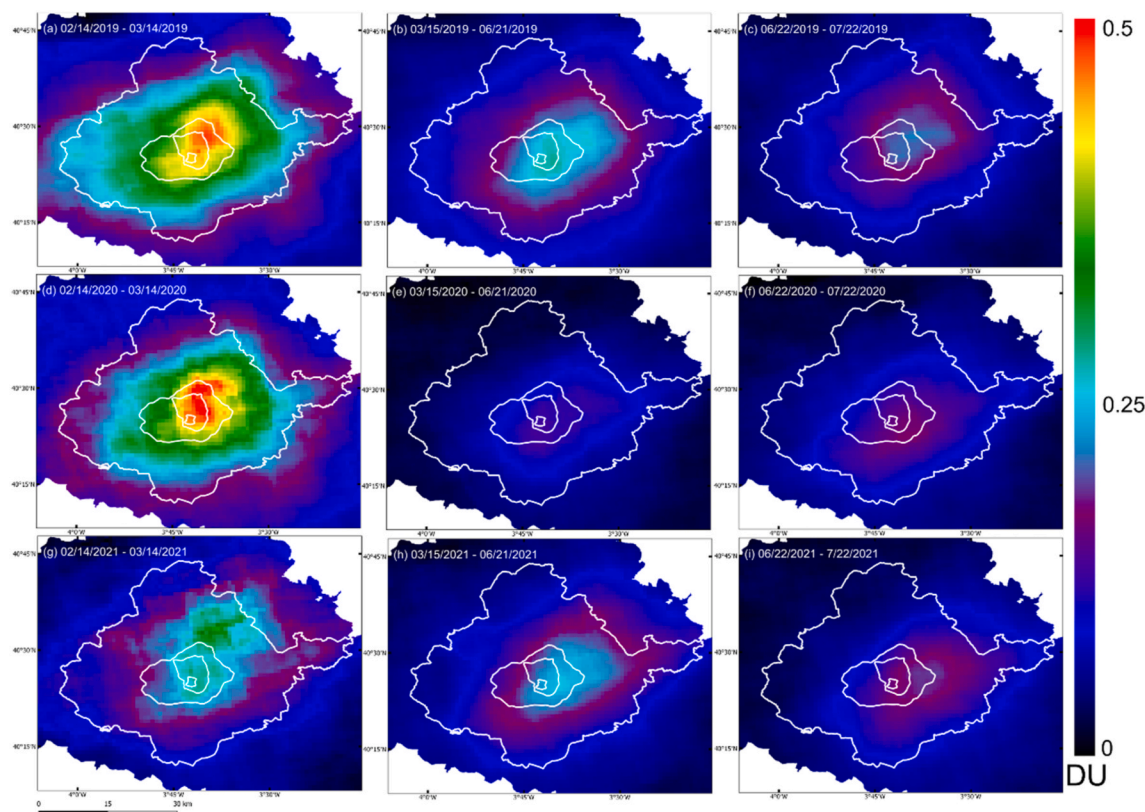


Fig. 10. Maps of mean Level-3 tropospheric TROPOMI NO₂ column (DU) between February 14th - March 14th, March 15th - June 21st, and June 22nd - July 22nd in (a, b, c) 2019, (d, e, f) 2020, and (g, h, i) 2021. The white lines from the inside to the outside represent the SPLEZ and the surrounding areas.

Table 3

Monthly average Level-3 tropospheric TROPOMI NO₂ concentration over the SPLEZ in 2022.

Month	Average NO ₂ concentration [DU]
January	0.50
February	0.55
March	0.24
April	0.19
May	0.22
June	0.14
July	0.16
August	0.15
September	0.17
October	0.32
November	0.39
December	0.31

the pandemic. These patterns are not exclusive to the SPLEZ, but their study enabled the characterization of the distribution of NO₂ from traffic. Fig. 10 shows that the spatial distribution has remained consistent across corresponding periods in different years. Fig. 10b–e, and h show the effects of 2020 mobility restrictions on NO₂ distribution and how pollution accumulates in northeastern areas outside the SPLEZ. During colder periods (Fig. 10a and d), domestic heating brings the SPLEZ once again closer to the epicenter of NO₂ concentration.

The analysis of the temporal evolution in 2022 further supports these findings. Table 3 and Fig. 11 show that winter is the period with the highest mean NO₂ concentration in the SPLEZ, especially in January (0.50 DU) and February (0.55 DU). The middle of the year is characterized by an overall lower pollution situation. Warmer months are particularly important for identifying the main sources of emissions, not only over the SPLEZ but also in the whole metropolitan area. The reduction in activities during summer vacations, coupled with the

increased presence of sunlight, contributes to a decrease in NO₂ pollution, and the remaining fraction can be mainly attributed to transportation. The magnitude of the enhancement in the NO₂ column observed by TROPOMI between April and May (Fig. 11) was different from surface-level measurements. This discrepancy could be attributed to meteorological conditions that usually affect the total number and quality of the available images (Schneider et al., 2021). Overall, the SPLEZ remains one of the main NO₂ hotspots in the city, as indicated by TROPOMI. This is particularly evident in colder periods. However, Level-3 images also show that NO₂ stemming from transportation does not play as much of a leading role in the area.

The results from data correlation are consistent with previous studies examining the relationship between NO₂ tropospheric column and surface mixing ratio (Cersosimo et al., 2020; Goldberg et al., 2021; Griffin et al., 2019; Jeong and Hong, 2021; Rudke et al., 2023). Rural sites generally show lower coefficients than urban locations. Higher correlations found in urban areas could also be attributed to a wider range of concentrations found over the year (as opposed to a small range in rural areas). Population density is also closely associated with NO₂ pollution (Sharifi and Felegari, 2022). Although the spatial resolution is still limited for assessing small-scale events (Goldberg et al., 2021), satellite imagery provides an accurate view of how NO₂ pollution distributes in peak hours of the day.

5. Conclusions

The impact of the implementation of the SPLEZ in the city of Madrid was evaluated using TROPOMI data. To facilitate the analysis, and due to the absence of a more extensive time series for comparison, three case studies were conducted. The first case compared the periods before and after the implementation of the SPLEZ. The second case analyzed the effect of an exceptional event, such as the COVID-19 pandemic, on anthropogenic emissions in the area. Finally, the third case involved

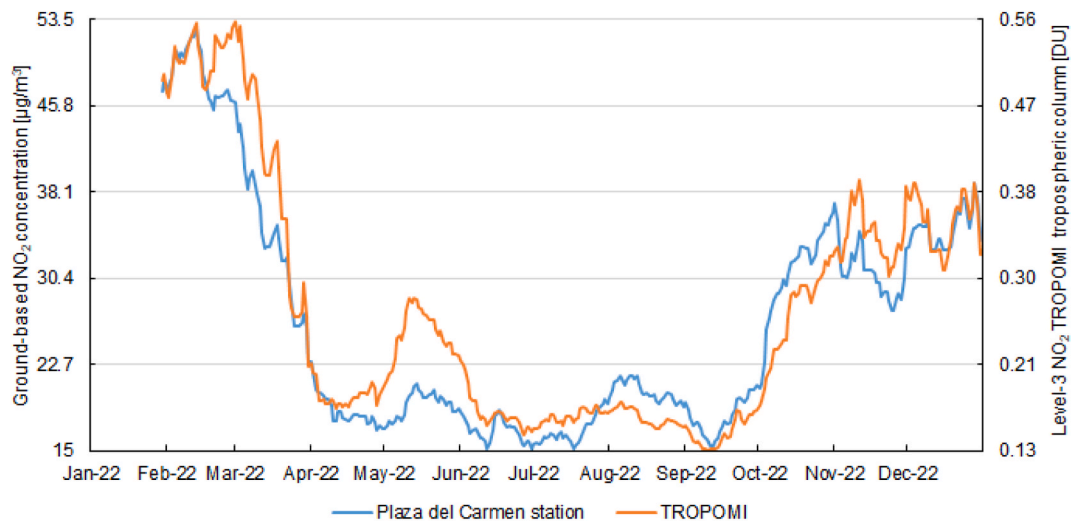


Fig. 11. 30-day moving average of (a) Level-3 tropospheric TROPOMI NO₂ concentration (DU) and (b) in situ NO₂ concentration (µg/m³) at Plaza del Carmen station location between 12:00 and 4:00 p.m. in 2022.

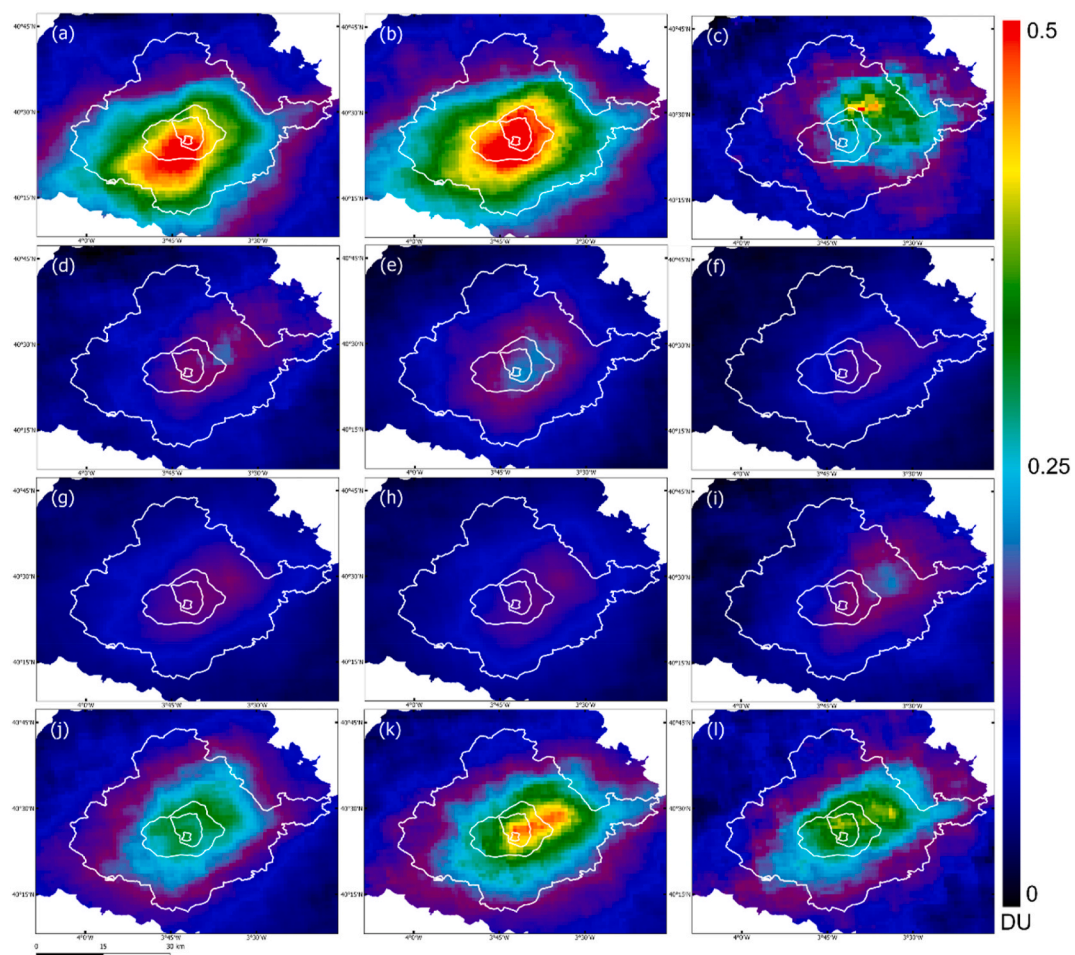


Fig. 12. Monthly maps of mean Level-3 tropospheric TROPOMI NO₂ column (DU) over the SPLEZ and its surroundings in 2022: (a) January, (b) February, (c) March, (d) April, (e) May, (f) June, (g) July, (h) August, (i) September, (j) October, (k) November and (l) December. The white lines from the inside to the outside represent the SPLEZ and the surrounding areas.

examining data from the most recent available full year (2022).

According to the seasonal analysis, during the winter, the use of fossil fuel heating positions the SPLEZ at the epicenter of NO₂ pollution. In warmer months, when road transportation becomes the main activity,

peak values are primarily directed toward northeastern territories. However, various environmental factors, especially high wind speeds and consistent wind directions, can definitely influence its distribution, complicating the comparisons with in situ measurements.

The results from the COVID-19 pandemic case study highlight the impact of human activities on the NO₂ concentration in both the SPLEZ and the entire metropolitan area. During the strict lockdown period, pollution reached its minimum levels in the SPLEZ. As certain activities resumed, emissions began to increase again. It is important to note that pollution levels have not yet returned to pre-pandemic levels in subsequent years. This can be attributed not only to meteorological factors but also to the lasting impact of the crisis and the measures that influenced the population's way of life during the new normality.

Previous research relying exclusively on air quality stations data found that high levels of NO₂ in the study area decreased following the implementation of the low-emission policy. This led to pollution levels falling below those in adjacent regions for the first time (Salas et al., 2021). This study complements these findings, suggesting that since the establishment of the SPLEZ, and despite the changes in the traffic restrictions criteria, NO₂ from road transport still plays a significant role in the zone, albeit to a lesser extent than in other regions of the metropolitan area. Remote sensing can detect distribution patterns and spatial differences similar to those observed by ground stations, especially over high pollution thresholds, although not without limitations.

It is necessary to consider that, although NO₂ mainly accumulates in the tropospheric layers, TROPOMI observes the entire atmospheric column, whereas ground-based stations provide surface mixing ratio data. Furthermore, despite its enhanced spatial resolution, the sensor may still face challenges in detecting spatial gradients at a small scale, especially with only one or two daily observations. However, it may be appropriate to allow a descriptive analysis of how emissions distribution can change as a result of policies and regulations, such as the implementation of LEZs. Satellite results were subjected to comparison using in situ data from the air quality stations network, yielding a linear correlation coefficient of 0.72, which is consistent with previous research. This confirms its suitability as a complement to ground level information for assessing the impact of low-emission policies.

CRedit authorship contribution statement

Carlos Morillas: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft. **Sergio Alvarez:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Project administration, Resources, Software, Visualization, Writing – review & editing. **Jose C.M. Pires:** Investigation, Methodology, Supervision, Validation, Writing – review & editing. **Adrian Jesus Garcia:** Investigation, Methodology, Software, Supervision, Writing – review & editing. **Sara Martinez:** Conceptualization, Data curation, Formal analysis, Project administration, Resources, Validation, Visualization, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.atmosenv.2024.120326>.

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